

Clustering

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Dependencies

This document depends on the following packages:

```
library(devtools)
library(Biobase)
library(dendextend)
```

To install these packages you can use the code (or if you are compiling the document, remove the `eval=FALSE` from the chunk.)

```
install.packages(c("devtools","dendextend"))
source("http://www.bioconductor.org/biocLite.R")
biocLite(c("Biobase"))
```

General principles

- How do we define close?
- How do we group things?
- How do we visualize the grouping?
- How do we interpret the grouping?

Load some data

We will use this expression set to look at how we use plots and tables to check for different characteristics

```
con =url("http://bowtie-bio.sourceforge.net/recount/ExpressionSets/bodymap_eset.RData")
load(file=con)
close(con)
bm = bodymap.eset
pdata=pData(bm)
edata=as.data.frame(exprs(bm))
fdata = fData(bm)
ls()
```

```
## [1] "bm"           "bodymap.eset" "con"           "edata"
## [5] "fdata"        "pdata"        "tropical"
```

Calculate distances between samples

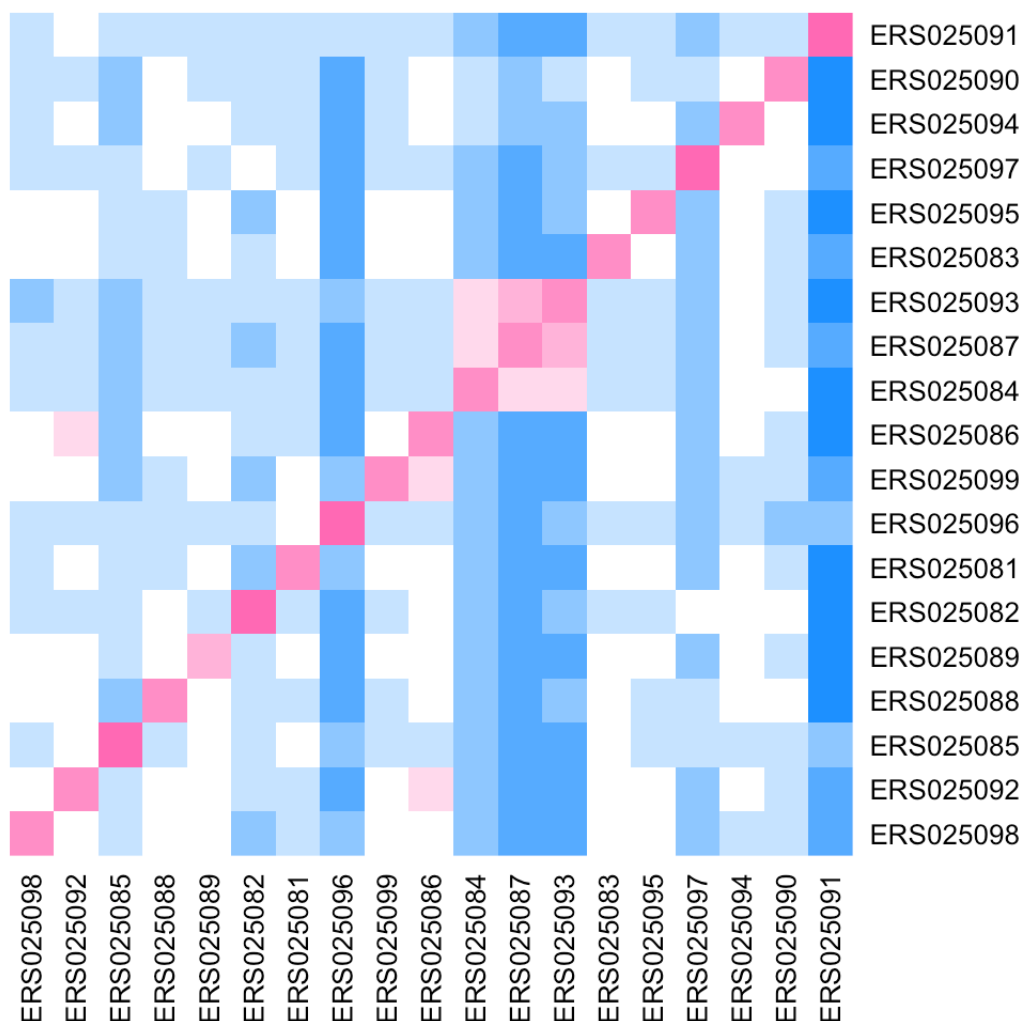
- Most important step
- Garbage in -> garbage out
- Distance or similarity
- Continuous - euclidean distance
- Continuous - correlation similarity
- Binary - manhattan distance
- Pick a distance/similarity that makes sense for your problem

First we log transform and remove lowly expressed genes, then calculate Euclidean distance

```
edata = edata[rowMeans(edata) > 5000,]
edata = log2(edata + 1)

# By default calculates the distance between rows
dist1 = dist(t(edata))

## Look at distance matrix
colramp = colorRampPalette(c(3,"white",2))(9)
heatmap(as.matrix(dist1),col=colramp,Colv=NA,Rowv=NA)
```

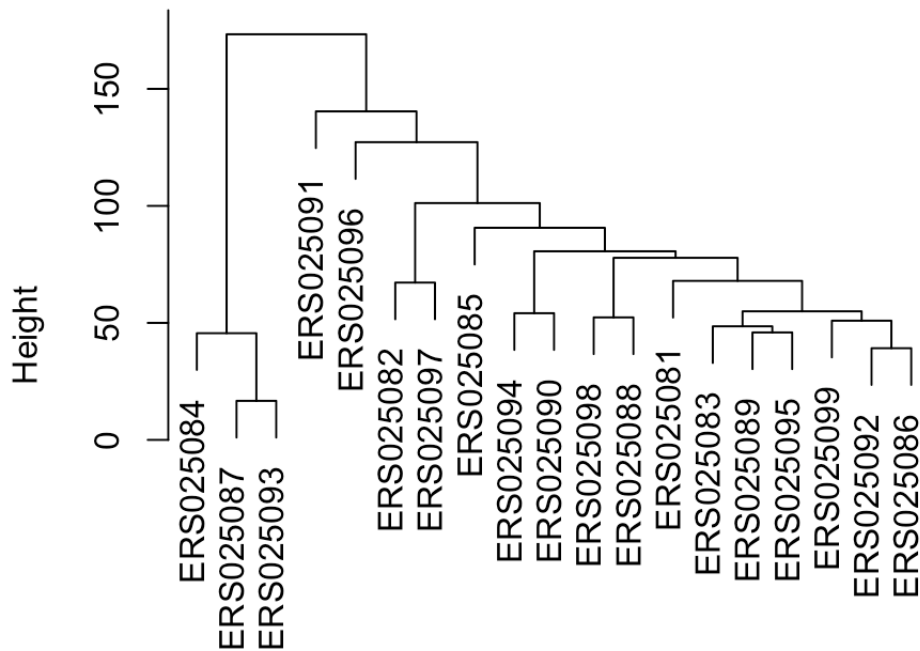


Now cluster the samples

Here we use the distance we previously calculated to perform a hierarchical clustering and plot the dendrogram:

```
hclust1 = hclust(dist1)
plot(hclust1)
```

Cluster Dendrogram

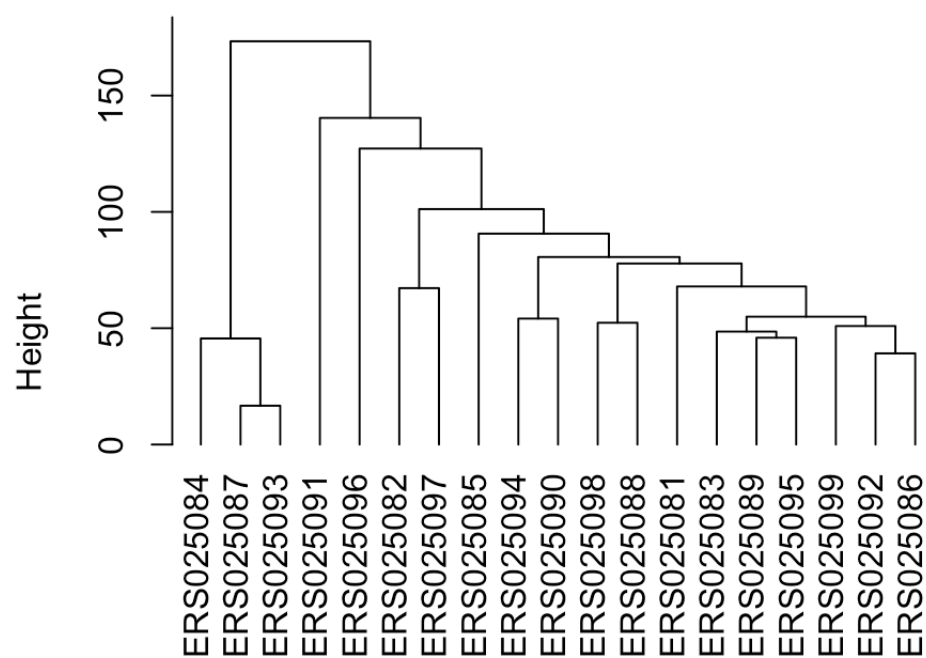


dist1
hclust (*, "complete")

We can also force all of the leaves to terminate at the same spot

```
plot(hclust1, hang=-1)
```

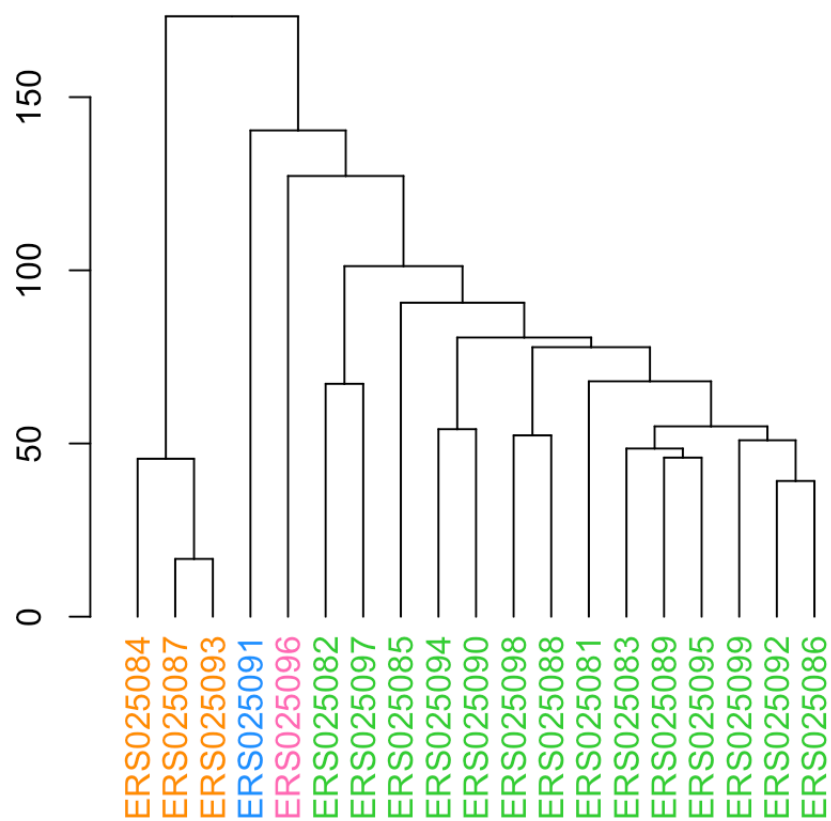
Cluster Dendrogram



dist1
hclust (*, "complete")

We can also color the dendrogram either into a fixed number of groups

```
dend = as.dendrogram(hclust1)
dend = color_labels(hclust1,4,col=1:4)
plot(dend)
```



Or you can color them directly

```
labels_colors(dend) = c(rep(1,10),rep(2,9))  
plot(dend)
```